

Linear differential equations with constant coefficients.

Lecture 4
DS

$$\mathcal{L}: C^n(\mathbb{R}) \rightarrow C(\mathbb{R})$$

$$x \in C^n(\mathbb{R}) \mapsto \mathcal{L}(x) = x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x$$

where $n \in \mathbb{N}^+$

Notation: $\mathcal{L}(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_n$

the characteristic polynomial

In order to find the general solution of the homogeneous LODE $\mathcal{L}(x) = 0$ we have the characteristic equation method.

We need $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ the roots of the char. pol.

$$\mathcal{L}(\lambda) \Rightarrow \boxed{\mathcal{L}(\lambda) = (\lambda - \lambda_1) \dots (\lambda - \lambda_m)}$$

• if $\lambda_1 = \alpha \in \mathbb{R}$ a root of multiplicity $m \geq 1$
 $\mapsto e^{\alpha t}, t \cdot e^{\alpha t}, \dots, t^{m-1} \cdot e^{\alpha t}$

• if $\lambda_{1,2} = \alpha \pm i\beta$ root of multiplicity $m \geq 1$
($\alpha, \beta \in \mathbb{R}, \beta > 0$)

$$\mapsto e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t, t \cdot e^{\alpha t} \cos \beta t, \dots, t^{m-1} \cdot e^{\alpha t} \sin \beta t \quad (\underline{2m} \text{ functions}).$$

We obtain $x_1, \dots, x_m$ (m functions)

• more stuff with D.

For $(x, y) \in \mathbb{R}^2 \setminus \{(0,0)\}$:

$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases}, \quad \rho = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

The gen. sol. of the n^{th} order LHDE $\mathcal{L}(x)=0$ is
 $x = c_1 x_1 + \dots + c_n x_n, \quad c_1, \dots, c_n \in \mathbb{R}.$

We consider now the Lm-HDE

$$(1) \quad \mathcal{L}(x) = f(t)$$

↳ non-homog. part.

Here $f \in C(\mathbb{R})$ is called the non-homog. part. of (1). or a force (physics)

Fundamental Theorem: (Th 2 from Lec 3)

The general solution of (1) is $x = x_h + x_p$,
where x_h is the general sol. of $\mathcal{L}(x) = 0$,
and x_p is a particular sol. of (1).

? How to find x_p ?

↳ Answer: There exists the Lagrange method
(the variation of constants method) for any Lm-HDE (of any order), (for any coeff.)

We presented this method in the seminar only in the case $m=1$.

"Property": Assume $\mathcal{L}$ has constant coefficients.
Assume f has such a form that
 $\exists$ a linear homog. diff. eq. w cc.
↳ another

whose $\tilde{f}$ is a solution.

Then eg. $\mathcal{L}(x) = f$ has a particular sol. x_p which has the same form as f .

Ex 7, Lab 2.

Lemma 1. • Let $f \in \mathbb{C}$. Then $(\underline{D} - \kappa_1)(e^{ft}) =$
 $\kappa_1 \in \mathbb{C}$
 $= f \cdot e^{ft} - \kappa_1 \cdot e^{ft}$
 $= e^{ft} \cdot (f - \kappa_1)$

• Let $k \in \mathbb{N}^*$. Then $(\underline{D} - \kappa_1) \cdot (t^k \cdot e^{ft}) =$
 $= k \cdot t^{k-1} \cdot e^{ft} + (f - \kappa_1) \cdot t^k \cdot e^{ft}$

• Let p be a polynomial in t and $\alpha \in \mathbb{R}$.

Then $\mathcal{L}(p(t) \cdot e^{\alpha t}) =$ "linear comb. of powers of t ."
 $= Q(t) \cdot e^{\alpha t}$

$\mathcal{L} = (\underline{D} - \kappa_1) \cdot \dots \cdot (\underline{D} - \kappa_m)$ "composition".

$Q \rightarrow$ polynomial in t with $\deg Q = \deg P - m$,
 for some $m \in \mathbb{N}$. Moreover, $m = 0$ when α
 is not a root of the char. pol. $l(\kappa)$ and
 $m \geq 1$ is the multiplicity of the root α
 of the ch. pol. $l(\kappa)$.

• Let $p_1, p_2 \in \mathbb{R}$ and $\alpha, \beta \in \mathbb{R}$ with $\beta > 0$.

Let $k \in \mathbb{N}$. Then $\mathcal{L}(t^k \cdot (p_1 \cdot e^{\alpha t} \cdot \cos \beta t + p_2 \cdot e^{\alpha t} \cdot \sin \beta t)) = Q(t) \cdot e^{\alpha t} \cdot \cos \beta t + Q(t) \cdot e^{\alpha t} \cdot \sin \beta t =$

$= g_1 \cdot e^{\alpha t} \cdot \cos \beta t + g_2 \cdot e^{\alpha t} \cdot \sin \beta t$ where $g_1, g_2 \in \mathbb{R}$ and
 we assume that $\alpha + i\beta$ is not a root of the
 ch. pol. l .

Examples:

1) $x' + 2x = f(t)$

$x' + 2x = 0$ (homog. eq.)

$\lambda + 2 = 0$

$\lambda_1 = -2 \rightarrow e^{-2t}$

$x_h = c \cdot e^{-2t}, c \in \mathbb{R}.$

$l(\lambda) = \lambda + 2$ (char. polynomial).

$f(t)$ sol. of the LHDE with CC.

Assumpt: $\tilde{L}(x) = 0$

we take $\tilde{l}(\lambda)$ the ch. pol. of $\tilde{L}$

We consider another polynomial: $\tilde{l}(\lambda) = \frac{(\lambda+2)}{\cancel{\lambda+2}} \cdot \tilde{l}(\lambda).$

Apply the ch. eq. method to the ch. pol. $\tilde{l}(\lambda).$

$\Rightarrow$ we obtain: functions $\underbrace{x_1, x_2, \dots, x_m, \dots, x_{m+\tilde{m}}}_{e^{-2t}, x_2, \dots, x_{1+\tilde{m}}}$

Look for a part. sol. of the form:

$x_p = d_1 x_2 + \dots + d_{\tilde{m}} x_{1+\tilde{m}}$, where the
 coeff. $d_1, \dots, d_m \in \mathbb{R}$ are found imposing
 that $x_p' + 2x_p = f(t).$

where f is a solution.

$$x' + 2x = \underbrace{5 \cdot e^t + 7 \cdot t \cdot e^{-t} - e^{-2t}}_{f(t)}$$

$f(t)$ is a sol. of a LHDE with CC.

$\tilde{L}(x) = 0 \Rightarrow \tilde{L}(\kappa)$ has the roots:

$$\rho_1 = 1, \rho_2 = -2, \rho_3 = -1 \text{ double}$$

$$\Rightarrow \tilde{L}(\kappa) = (\kappa - 1)(\kappa + 2)(\kappa + 1)^2 \quad (\text{char. poly.})$$

$$\begin{aligned} \tilde{L}(\kappa) &= (\kappa + 2)(\kappa - 1)(\kappa + 2)(\kappa + 1)^2 \\ &= (\kappa + 2)^2(\kappa - 1)(\kappa + 1)^2 \end{aligned}$$

-2 ~~simple~~ double sol. $\rightarrow e^{-2t}, t e^{-2t}$

1 simple sol. $\rightarrow e^t$

-1 double sol. $\rightarrow e^{-t}, t \cdot e^{-t}$

lim. indep.

$$\text{Search } x_p = \underline{d_1} \cdot t \cdot e^{-2t} + \underline{d_2} \cdot e^t + \underline{d_3} \cdot e^{-t} + \underline{d_4} \cdot t \cdot e^{-t}$$

$$d_1, d_2, d_3, d_4 = ?$$

$$\boxed{x_p' + 2x_p = f(t)}$$

$$\begin{aligned} \mathcal{L}(d \cdot e^t) &= (d \cdot e^t)' + 2d \cdot e^t = d \cdot e^t + 2d \cdot e^t \\ &= 3d \cdot e^t \end{aligned}$$

$$\begin{aligned} x_p &= \underline{d_1} \cdot e^{-2t} - \cancel{2d_1 \cdot t \cdot e^{-2t}} + d_2 \cdot e^t - d_3 \cdot e^{-t} + \\ &+ d_4 \cdot e^{-t} - d_4 \cdot t \cdot e^{-t} + \cancel{2d_1 \cdot t \cdot e^{-2t}} + \\ &+ 2d_2 \cdot e^t + 2d_3 \cdot e^{-t} + 2d_4 \cdot t \cdot e^{-t} \end{aligned}$$

$$\Rightarrow 5 \cdot e^t + 7 \cdot t \cdot e^{-t} - \underline{e^{-2t}}, t \in \mathbb{R}.$$

$$\Rightarrow \underline{d_1 = -1}$$

$$\Rightarrow t \cdot e^{-2t} : 0$$

~~e^{-2t}~~ :

$$\Rightarrow e^t : d_2 + 2d_1 = 5 \Rightarrow \underline{d_2 = \frac{5}{3}}.$$

$$\Rightarrow e^{-t} : -d_3 + d_4 + 2d_3 = 0$$

$$\Rightarrow d_4 = -d_3 \Rightarrow \underline{\frac{5}{3}} \Rightarrow d_3 = -\frac{5}{3}$$

$$\Rightarrow t \cdot e^{-t} \Rightarrow -d_4 + 2d_4 = 7$$

$$\Rightarrow d_4 = 7.$$



$$x_p = -t \cdot e^{-2t} + \frac{5}{3} \cdot e^t - 7 \cdot e^{-t} + 7 \cdot t \cdot e^{-t}.$$

$$2) x' + 2x = t^3.$$

$$f(t) = t^3.$$

? $\exists$ a l.h.d.e w cc, $\tilde{L}(x) = 0$ s.t. f is a sol. of it?

$\hookrightarrow$ yes, when $f = 0$ is a root of the ch. pol.

$$\tilde{L}(x) = x^4$$

Consider $\tilde{L}(x) = (x+2) \cdot x^4$

$$\hookrightarrow \boxed{e^{-2t}}, 1, t, t^2, t^3$$

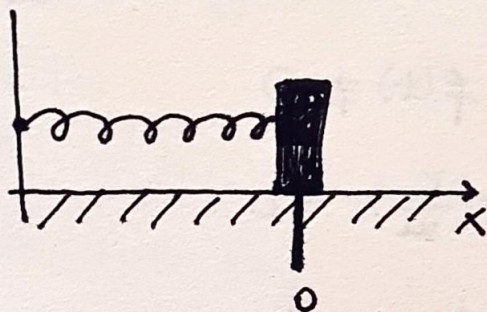
$\downarrow$
taken
already
(homog.)

where f is a solution.

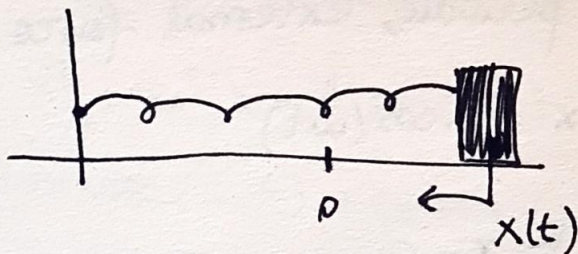
$$x_p = d_1 + d_2 \cdot t + d_3 \cdot t^2 + d_4 \cdot t^3$$

$$d_1, d_2, d_3, d_4 = ? \quad x_p' + 2x_p = t^3. \quad \dots$$

The spring-mass system



$x(t)$



$m > 0$ is the mass

The second Newton's law:

$$m \cdot x''(t) = -F_x - F_f + f(t)$$

$F_x \rightarrow$ the restoring force

$F_f \rightarrow$ the friction

$f \rightarrow$ the external force.

Hooke's law: $F_x = K \cdot x$, $K > 0$

$$F_f = \gamma \cdot x', \quad \gamma \geq 0$$

damping
coeff

H

... is it periodic?

$$m \cdot x'' = -k \cdot x - \gamma \cdot x' + f(t) \quad | : m$$

$$\Rightarrow x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = f(t)$$

external force.

Case 1: $\gamma = 0, f(t) \equiv 0$

un-damped motion, without external force.

Case 2: $\gamma > 0, f(t) \equiv 0$

$$x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = 0$$

Case 3: $\gamma = 0, f(t) = A \cdot \cos(\omega t)$

bounded, periodic, external force.

$$x'' + \frac{k}{m} \cdot x = A \cdot \cos(\omega t)$$

$\rightarrow$ Case 1: $x'' + \frac{k}{m} x = 0$

$$\omega_0 = \sqrt{\frac{k}{m}} > 0$$

$$x'' + \omega_0^2 \cdot x = 0$$

$$x^2 + \omega_0^2 = 0$$

$$x^2 = -\omega_0^2 < 0$$

$$x_{1,2} = \pm i \cdot \omega_0$$

$\rightarrow \cos(\omega_0 t), \sin(\omega_0 t).$

$$x = c_1 \cdot \cos(\omega_0 t) + c_2 \cdot \sin(\omega_0 t) \quad (1)$$

any function is $\frac{2\pi}{\omega_0} \rightarrow$ periodic.

whose f is a solution.

oscillatory (with constant amplitude)

For $(c_1, c_2) \in \mathbb{R}^2 \setminus \{(0,0)\}$ we take the polar coord:

$$\begin{cases} c_1 = A_0 \cdot \cos \rho_0 \\ c_2 = A_0 \cdot \sin \rho_0 \end{cases} \quad \begin{aligned} A_0 &= \sqrt{c_1^2 + c_2^2} \\ \tan \rho_0 &= \frac{c_2}{c_1} = \frac{y}{x} \end{aligned}$$

Introduce in (1):

$$\begin{aligned} X &= A_0 \cdot \cos \rho_0 \cdot \cos(\omega t) + A_0 \cdot \sin \rho_0 \cdot \sin(\omega t) \\ &= A_0 (\cos(\omega_0 t - \rho_0)) \end{aligned}$$

↓

amplitude of the oscillations.

$$\rightarrow \text{Case 3: } x'' + \omega_0^2 x = A \cdot \cos(\omega t)$$

$$x_h = c_1 \cdot \cos(\omega_0 t) + c_2 \cdot \sin(\omega_0 t), \quad c_1, c_2 \in \mathbb{R}.$$

$$\boxed{X = x_h + x_p} \quad x_p = ?$$

$$\rightarrow \text{Case 3.1: } \omega \neq \omega_0$$

$$\begin{aligned} x_p &= \underline{d_1} \cdot \cos(\omega t) + \underline{d_2} \cdot \sin(\omega t) \\ d_1, d_2 &= ? \end{aligned}$$

$$\begin{aligned} x &= \underline{c_1} \cdot \cos(\omega_0 t) + \underline{c_2} \cdot \sin(\omega_0 t) + x_p \\ c_1, c_2 &\in \mathbb{R}. \end{aligned}$$

(bounded solution because it's a sum of $\sin, \cos$; oscillatory)

$$\text{HW: } \begin{cases} \cos t - \cos 2t & \text{periodic?} \\ \cos t - \cos \sqrt{2}t & \text{periodic?} \end{cases}$$

→ Case 3.2 $\omega = \omega_0$.

$$x_p = t(\underline{d_1} \cdot \cos(\omega_0 \cdot t) + \underline{d_2} \cdot \sin(\omega_0 \cdot t))$$

$$d_1, d_2 = ?$$

↳ not bounded sol. (if we add it to a bounded one $\Rightarrow$ each one is unbounded) $\Rightarrow$ RESONANCE.

$\Rightarrow$ (†) solution is unbounded, although external force is bounded.